ROYAL GLOBAL UNIVERSITY

GUWAHATI

ROYAL SCHOOL OF BIO SCIENCES

(RSBSC)

DEPARTMENT OF BIOCHEMISTRY

Course Structure and Syllabus

Based on National Education Policy -2020

FOR

M.Sc. Biochemistry

2 Year Single Major

W.E.F. 2025-26

W.E.F.2025-26

STRUCTURE OF THE SYLLABUS FOR 2 YEAR PG PROGRAMME

SCHOOL NAME - **Royal School of Biosciences (RSBSC)**

DEPARTMENT NAME - **Department of Biochemistry**

PROGRAMME NAME - **Postgraduate programme (PG)**

1st SEMESTER				
COURSE CODE	**COURSE TITLE**	**LEVEL**	**CREDIT**	**L-T-P**
BCH154M101	Chemistry of Biomolecules	400	4	4-0-0
BCH154M102	Cell Biology	400	4	4-0-0
BCH154M103	Bioenergetics & Metabolism	400	4	4-0-0
BCH154M104	Bioanalytical technique	400	4	4-0-0
BCH154M114	Practical on Biomolecules and Cell Biology	400	4	0-0-8
MOOCs 1	*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online platform/SWAYAM portal	400	4	
		TOTAL CREDIT FOR 1st SEMESTER	**24**	16-0-8
2nd SEMESTER				
COURSE CODE	**COURSE TITLE**	**LEVEL**	**CREDIT**	**L-T-P**
BCH154M201	Enzymology	500	4	4-0-0
BCH154M202	Molecular Biology	500	4	4-0-0
BCH154M203	Immunology	500	4	4-0-0
BCH154M204	Genetics for biologics	500	4	4-0-0
BCH154M214	Practical on Enzymes, Molecular Biology & Immunology	500	4	0-0-8
MOOCs 1	*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online platform/SWAYAM portal	100	4	
		TOTAL CREDIT FOR 2nd SEMESTER	**24**	16-0-8
TOTAL CREDIT FOR 1st YEAR = 48				
3rd SEMESTER				
COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
BCH154M301	Advanced Enzymology	500	4	4-0-0
BCH154M302	Clinical Biochemistry	500	4	4-0-0
BCH154M303	Neurobiochemistry	500	4	4-0-0
BCH154M304	Nutritional Biochemistry	500	4	4-0-0
MIB154M304	Practical on Enzymes and Clinical Biochemistry	500	4	0-0-8
		TOTAL CREDIT FOR 3rd SEMESTER	**20**	16-0-8
OR 3rd SEMESTER (For students with 3rd and 4th Semester Research)				
BCH154R321	RESEARCH PROJECT – PHASE I	500	20	0-0-20
4th SEMESTER				
COURSE CODE	**COURSE TITLE**	**LEVEL**	**CREDIT**	**L-T-P**

BCH154P221	Dissertation (students with research in 4th Sem)	500	20	
(for 'Coursework only' in lieu of Research)				
BCH154M401	**Microbiology**	500	4	4-0-0
BCH154M402	**Membrane Biology**	500	4	4-0-0
BCH154P221	Dissertation–2 [One year PG (course work + research)]	500	12	
OR 4th SEMESTER (For students with 3rd and 4th Semester Research)				
BCH154R421	RESEARCH PROJECT– PHASE 2	500	**20**	
TOTAL CREDIT FOR 2nd YEAR = 40				

1ST SEMESTER SYLLABUS
PAPER I: CHEMISTRY OF BIOMOLECULES SUBJECT CODE: BCH154M101 CREDIT UNITS: L-T-P-C = 3-0-0-4 STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
The course is designed to introduce the analytical aspect of biochemistry and educate students about the various bioorganic and biochemical reactions and formulations.	**CO1**. Students will recall the core knowledge base in the theory and practice of modern Biochemistry and Biophysics. **CO2**. Students will understand the mechanism of different Biochemical and Biophysical reactions **CO3**. Students will be able to analyze the coordination necessary in biochemical functioning, like in the case of hemoglobin **CO4**. Students will be able to analyze the biochemistry of micro nutrients	Powerpoint presentations; Teaching using chalk and board; Oral discussion sessions in the Class	Oral questions will be asked in the class. Problems will be assigned to test student's Analytical ability. Class tests will be conducted for internal Assessment.

Detailed Syllabus:-

Modules	Course content	Periods
I	**Basics of Carbohydrate:** Stereochemistry, chirality, R.S. designation; chemistry and properties of water; pH and indicators; acid-base concept; mono- di- and poly-protic acids; buffer solutions and their action; reactive reaction intermediates (carbocation, carbanion, carbenecarbyne, and free radicals); dielectric constant; surface tension; viscosity; dipole moment. Classifications of Carbohydrates: Monosaccharides, Disaccharides, and Polysaccharides: Structural and functional concepts. Mutarotation of sugars. Key Reactions of Carbohydrates.	**12**
II	**Lipids:**Building blocks of lipids- fatty acids, glycerol, ceramide. Storage lipids- triacyl glycerol and waxes. Structural lipids in membranes– glycerophospholipids, galactolipids and sulpholipids, sphingolipids and sterols, structure, distribution and role of membrane lipids. Importance of Cholesterol.	**12**
III	**Structure and properties of Nucleic acids:**Basic concepts of Nucleotides and nucleosides. Structure of DNA: DNA double helix;Forces stabilizing the double helical structure, Tm and Cot curve for denaturation and renaturation of DNA. DNA as genetic material. RNA: Different types of RNA, biological functions of mRNA, rRNA and tRNA.	**12**
IV	**Amino Acids and Proteins:** Structure, properties, and classification of amino acids; titration curves; peptide bond & Ramachandran plot; Important reactions of amino acids. Levels of protein structure; globular, membrane & fibrous proteins; interactions stabilizing proteins; supersecondary& domain structure; protein targeting; protein folding, misfolding& aggregation; molecular chaperones; unstructured proteins; protein dynamics; naturally occurring peptides.Structural and functional aspects of haemoglobin and myoglobin: Oxygen binding property, oxy- and deoxy- forms, Relaxed and tense (R &T) configurations. Iron Containing Non-hemeMetalloproteins: transferrin and ferritin.	**12**
Total		**48**

Text Books:

1. Branden, C.I. and Tooze, J. (2009) Introduction to Protein Structure. Second Edition, C.R.C. Press.
2. Finkelstein, A.V. and Ptitsyn, O.B. (2016) Protein Physics: A course of lectures. Second Edition, Academic Press Publications.
3. Almeida, P. (2016) Proteins Concepts in Biochemistry. Garland Science Publishers.
4. Nelson D.L. and Cox M.M. (2017) Lehninger's Principles of Biochemistry, Seventh Edition, Freeman & Co, New York.
5. Atkins P.W. (2017) the Elements of Physical Chemistry.Seventh Edition, Oxford Univ. Press.
6. Finar I. L. (2012) Organic Chemistry, Pearsons.
7. Morrison R.T. and Boyd R.N. (2009) Organic Chemistry, Seventh Edition, Pearsons.
8. Stryer L. (2015) Biochemistry, Eight Editions. W.H. Freeman.
9. Voet, D. and Voet, J.G. (2011) Biochemistry. Fourth Edition, John Wiley & Sons.

Reference books:

1. Principles of physical chemistry- Puri,Sharma
2. Baxevanis A.D. And Ouellette B.F.F. (2001) Bioinformatics, Second edition, Wiley Interscience.
3. Mount D.W. (2004) Bioinformatics: Sequence and Genome Analysis, Second Edition. CSHL Press.
4. Lesk A.M. (2013) Introduction to Bioinformatics.Fourth Edition. Oxford University Press.
5. Grohima M.M. (2010) Protein Bioinformatics. Elsevier Publications.
6. Ghosh Z. and Mallick B. (2008) Bioinformatics: Principles and Applications. Oxford University Press.

PAPER II: CELL BIOLOGY
SUBJECT CODE: BCH154M102
CREDIT UNITS: L-T-P-C = 4-0-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
The course entails to educate the student about the basic biology and	**CO 1**. Students will recall the core knowledge of different structural and functional properties of a cell **CO 2**. Students will understand the role of compartmentalization and the importance of cell biology in biochemistry.	Chalk and board teachings, power point presentations, and video presentations on the step by step process	Discussions, oral Questioning and analytical questions will be given to

physiology of a cell, and important structural and functional properties	**CO 3**. Students will apply these principals to current biological questions of today. **CO 4**. Students will analyze how cells grow, divide, and die and the importance of these processes.	of mitosis and meiosis will be shown, chronology of molecular events in various processes will be shown as flowchart.	students. Internal exams and seminars will be conducted

Detailed Syllabus:-

Modules	Course content	Periods
I	**Basic concepts of cell biology:** Evolution of cells, cell architecture, structural organization of prokaryotic and eukaryotic cells. Structure and functions of different cell organelles: nucleus, mitochondria, chloroplast, rough and smooth endoplasmic reticulum, Golgi apparatus, lysosomes, and peroxisomes. Cytoskeleton: actin, microfilaments, intermediate filaments, and microtubules. Adherence junctions, tight junctions, gap junctions, desmosomes, hemidesmosomes, focal adhesions, and plasmodesmata. Subcellular fractionation and Svedberg unit.	**12**
II	**Chromosomal organization, cell cycle and cell division:** Definition of a gene, chromosomal organization of gene; Types of chromosomes: autosomes, sex chromosomes; Karyotyping; Chromatin, Histones and Nucelosomes;Heterochromatin and euchromatin;Structural chromosomal abnormalities: Chromosome ploidy, Polyploidy and aneuploidy. Cell cycle and its regulation; cell cycle checkpoints Events of mitosis and cytokinesis; Events of meiosis, Regulation of cell division.	**12**
III	**Intracellular trafficking and its importance in cell function:** Overview of vesicle-mediated transport pathways - exocytosis, endocytosis, and retrograde transport; Vesicle formation and cargo selection: adaptins, clathrin, and cargo receptors. Role of small GTPases (Rab, Rho etc.) in vesicle targeting and fusion; Protein sorting in the trans-Golgi network and endosomes; SNARE proteins and membrane fusion: v-SNAREs, t-SNAREs, and NSF-SNAP complex;Specificity in vesicle targeting and membrane fusion.	**12**
IV	**Signal Transduction pathways for cellular growth, development and death**: Types of cell signaling: autocrine, paracrine, and endocrine signaling; Cell surface receptors and intracellular receptors ; Mechanisms of ligand binding, receptor activation, and conformational changes. Introduction to second messengers like cAMP and calcium; Role of second messengers in signal transduction; Activation of G Protein-Coupled Receptors (GPCRs) and receptor tyrosine kinase (RTKs); signaling pathways Ras-MAPK, PI3K/Akt/mToR, JAK-STAT pathway. Mechanism of cell death pathways: Apoptosis, Necrosis, Pyroptosis, and Ferroptosis.The physiological importance of Autophagy and senescence.	**12**
Total		**48**

Text Books:

1. Molecular Biology of the Cell, 6th Edition. Bruce Alberts, Dennis Bray, JulianLewis,
2. Martin Raff, Keith Roberts, and James D Watson. Publisher New York: GarlandScience.
3. The Cell: A Molecular Approach, 7th Edition, by Geoffrey M. Cooper and RobertE. Hausman, published by ASMPress.
4. *Lehninger Principles of Biochemistry,* Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, U.S.A. 7th edition, 2017
5. Molecular Cell Biology; Lodish et al., 8thEdn. W.H. Freeman and Co. (2012).
6. Biochemistry 8th Edn. Jeremy M. Berg, John L. Tymoczko, LubertStryer.
7. Victor Rodwell, David Bender, P. Anthony Weil, Peter Kennelly. *Harpers Illustrated Biochemistry*

31th Edition,2018

Reference Books:

1. Lipid Biochemistry; 5th Edn. Michael I. Gurr, John L. Harwood and Keith N. Frayn, Blackwell Science (2002).
2. Principles of Human Physiology; 6th Edn. Cindy L. Stanfield Pearson,(2011).
3. Biochemistry Ed. Donald Voet& Judith G. Voet, 4thEdn. John Wiley & Sons, Inc. (2012).
4. Mammalian Biochemistry; White, Handler and Smith, McGraw-Hill,(1986)

PAPER III: BIOENERGETICS AND METABOLISM
SUBJECT CODE: BCH154M103
CREDIT UNITS: L-T-P-C = 3-1-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	**Course Learning Outcomes**	**Teaching and Learning Activity**	**Assessment Tasks**
The objective of the course is to introduce the students with fundamental concepts of energy production and metabolism of the biological macromolecules	**CO1.** Students will remember the thermodynamic parameters associated with the metabolism of biological macromolecules. **CO2.** Students will understand the role of different metabolic enzymes in biological catalysis **CO3.** Student will define biochemical functions and integrated metabolism in different tissues **CO4.** Students will analyze the key regulatory points in metabolic pathways	Chalk and board teachings, power point presentations	Discussions,oral Questioning and analytical questions involving calculations of bioenergetics will be given to students. Internal exams and seminars will be conducted.

Detailed Syllabus:-

Modules	**Course content**	**Periods**
I	**Bioenergetics:** Energy transformation, Laws of thermodynamics, Biological oxidations, oxygenases, hydroxylases, dehydrogenases, and energy transducing membranes. Gibbs energy, free energy changes, and redox potentials, phosphate potential, ion electrochemicalpotentials,proton electrochemical potential, membrane potentials, photons energy interconversions. Ion transport across energy-transducingmembranes. The influxandeffluxmechanisms.Proton circuit and electrochemical gradient, the transport andDistribution of actions, anions, and ionophores. Uniport, antiport, and symport mechanisms, shuttle systems. The mitochondrial respiratory chain, order and organization of carriers, proton gradient, iron-sulfur proteins, cytochromes, and their characterization.	**12**
II	**Intermediary metabolism:** Approaches for studying metabolism. Carbohydrates: Glycolysis, citric acid cycle, its function in energy generation and biosynthesis of energy-rich bonds,pentose phosphate pathway, anditsregulation.Alternate pathways of carbohydrate metabolism. Gluconeogenesis, interconversions of sugars. Biosynthesis of glycogen, starch, and oligosaccharides. Regulation of blood glucose	**12**

	homeostasis. Hormonal regulation of carbohydratemetabolism.	
III	**Lipids metabolism:**Lipids Fatty acid biosynthesis: Acetyl CoA carboxylase, Fatty acid synthase, desaturase, and elongase. Fatty acid oxidationand its regulation. Lipid Biosynthesis: Biosynthesis of triacylglycerols, phosphoglycerides, and sphingolipids,Biosyntheticpathwaysforterpenes,steroids, and prostaglandins. Ketone bodies: Formation and utilization. Metabolism of Circulating lipids: chylomicrons, LDL, HDL, and VLDL. Free fatty acids. Lipid levels in pathological conditions.	**12**
IV	**Metabolism of amino acids and nucleotides:**Amino Acids biosynthesis and degradation. Urea cycleanditsregulation,Nucleicacidsbiosynthesisorpurines and pyrimidines Degradation of purines and pyrimidines. Regulation of purine and pyrimidine biosynthesis Structure and regulation of ribonucleotide reductase. Biosynthesis of ribonucleotides, deoxyribonucleotides, and polynucleotides. Inhibitors of nucleic acidbiosynthesis	**12**
Total		**48**

Text Books:

1. *Lehninger Principles of Biochemistry*, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, U.S.A. 7th edition, 2017
2. Harper's Illustrated Biochemistry; 31thEdn. Robert K. Murray, Daryl K. Granner,Victor
3. W. Rodwell, the McGraw-Hill (2018).
4. Biochemistry by Donald Voet and Judith Voet, John Wiley and Sons NY. 4thedtn, 2012.

Reference Books:

1. Berg, J. M., Tymoczko, J. L. and Stryer. *Biochemistry*,, W.H Freemanand Co., 9thEdition, 2019.
2. Buchanan, B., Gruissem, W. and Jones, R.,*BiochemistryandMolecularBiologyofPlants*, 2ndEdn, American Society of Plant Biologists,U.S.A.
3. Fundamentals of Biochemistry by J.L.Jain, Nitin Jain and Sunjay Jain, S.ChandGroup.

PAPER IV: BIOANALYTICAL TECHNIQUES
Subject Code:BCH154M104
Credits: 3-1-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	**Course Learning Outcomes**	**Teaching and Learning Activity**	**Assessment Tasks**
The objective of the course is to provide students with a broad understanding of the principles of bioanalytical	**CO1.** Students willgain familiarity with working principle, tools and methodology of analytical techniques **CO2.** Students will understand the strengths, limitations and creative use of techniques in biological science **CO3.** Students will execute the techniques for biological experiments	Marker and board, Powerpoint presentations, student interaction	Students will be given assignment for example; oral questions will be asked

instrumentation and to provide an appreciation of their uses.	**CO4**. Students will be able to make a strategy on molecular techniques for the improvement of the results		

Detailed Syllabus:-

Modules	Course content	Periods
I	Electrophoretic techniques: Principles of electrophoretic separation. Continuous, free, zonal and capillary electrophoresis, different types of electrophoresis including paper, cellulose, and pulse field gel electrophoresis.Spectroscopy: Concepts of spectroscopy, Visible and U.V. spectroscopy, Laws of photometry.Beer-Lambert's law, Principles and applications of colorimetry.	12
II	Chromatography: Principles of partition chromatography, paper, thin layer, ion exchange and affinity chromatography, gel permeation chromatography, HPLC, gas chromatography.	12
III	Centrifugation: Principles of centrifugation, concepts of R.C.F., different types of instruments and rotors, preparative, differential and density gradient centrifugation, analytical ultra-centrifugation, determination of molecular weights and other applications, sub cellular fractionation	12
IV	Electron microscopy: Light, electron (scanning and transmission), phase contrast, fluorescence microscopy, freeze-fracture techniques, specific staining of organelles or marker enzymes.	12
Total		**48**

Text books:

1. Principles and Techniques of Practical Biochemistry, Keith Wilson and John Walker, 5thedition, 2000.
2. Physical Biochemistry, application to Biochemistry and Molecular Biology, David Freifelder, 2nd Edition, 1982.

Reference books:

1. R. F. Boyer, Modern experimental biochemistry, Benjamin Cummings, San Francisco, 3rd ed., 2000.
2. R. F. Boyer, Biochemistry laboratory: modern theory and techniques, Prentice Hall, Boston, 2nd ed., 2012.
3. D. Harvey, Modern analytical chemistry, McGraw-Hill, Boston, 2000.

PAPER IV: Practical on Biomolecules and Cell Biology
SUBJECT CODE: BCH154M114
CREDIT UNITS: L-T-P-C = 0-0-8-4
STUDENT'S SCHEME OF EVALUATION: PRACTICAL (P)

Facilitating the achievement of Course Learning Outcomes

Course Objective	**Course Learning Outcomes**	**Teaching and Learning Activity**	**Assessment Tasks**
The objective of the course is to introduce the students with	**CO1.** Students will remember the thermodynamic parameters associated with the metabolism of biological macromolecules. **CO2.** Students will understand the role of	Chalk and board teachings, video presentations of the methology,	Discussions, oral Questioning and analytical questions will be

fundamental concepts of energy production and metabolism of the biological macromolecules	different metabolic enzymes in biological catalysis **CO3.** Student will define biochemical functions and integrated metabolism in different tissues **CO4.** Students will analyze the key regulatory points in metabolic pathways	calculations guidance, conversion of moles, molar, mg, g, L etc, weighing, handling equipments and hands on practicals	given to students. Students' hands on' learning will be regularly monitored

Detailed Syllabus:-

Modules	Course content	Periods
I	1. Preparation of buffer using Henderson-Hasselbach equation and determination of its buffering capacity by acid and alkali. 2. Estimation of serum inorganic phosphate. 3. Formal titration of amino acids.	24
II	1. Estimation of D.N.A. using diphenylamine. 2. Estimation of R.N.A. usingorcinol. 3. Determination of free proline.	24
III	1. Estimation of Carbohydrate by Anthronemethods. 2. Estimation of protein by Lowry's method. 3. Microscopic examination ofurine. 4. Microscopic examination and chemical analyses ofblood.	24
IV	1. Separation of amino acid mixtures by paper chromatography. 2. Separation of lipids by T.L.C. 3. Molecular docking study of protein and drugs using M.G.L. tools.	24
Total		**96**

Text books:

1. Practical Biochemistry – 3rdEdn. DavidPlummer.
2. Introductory Practical Biochemistry – S.K.Sawhney and Randhir Singh.

Reference book:

1. Practical Clinical Biochemistry, ed. Harold Varley, 4th edn. C.B.S.Publishers.
2. Practical Clinical Biochemistry: Methods and Interpretation, 4th edn. RanjnaChawla, Jaypee Brothers Medical Publishers.
3. Practical and Clinical Biochemistry for Medical Students, ed. T.N.Pattabhiraman, Gajanna Publishers.

2^{ND} SEMESTER SYLLABUS
PAPER I: ENZYMOLOGY SUBJECT CODE: BCH154M201 CREDIT UNITS: L-T-P-C = 3-1-0-4 STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	**Course Learning Outcomes**	**Teaching and Learning Activity**	**Assessment Tasks**
To introduce the students to enzyme types, mechanisms and regulation.	Students will learn about **CO1**. Students will learn about Different enzyme classifications, various aspects of action mechanism of enzyme, role of metals in enzyme action, organization of enzymes in cells and enzyme mechanisms and regulation. **CO2.** Students will understand the basis of enzyme classification, role of various factors in enzyme activity, role of amino acids in enzyme actions and the role of cellular conditions in enzyme regulation. **CO3**. Students will be able to analyse the role of different amino acids and cofactors in enzyme reactions. **CO4.** Student will be able to evaluate the important amino acids in enzyme reactions	Chalk and board teachings, powerpoint presentations	Discussions, oral Questioning and analytical questions will be given to students. Internal exams and seminars will be conducted

Detailed Syllabus:-

Modules	**Course content**	**Periods**
I	**Introduction to enzymes:** Enzyme nomenclature and classification, Isozymes; Ribozymes; Zymogens; Abzymes; Structure and General properties of enzymes; Active site and Specificity of enzyme; Enzyme substrate complex. Induced fit theory. Mechanism of enzyme action, Factors affecting enzyme activity; Isozymes; Coenzymes, Metalloenzymes; membrane bound enzymes; Multienzyme complexes. Thermodynamics and Equilibrium; Enzyme activity; Specific activity and Units; Classification and nomenclature of enzymes.Enzyme assays: Types, Continuous and discontinuous assays.	12
II	**Enzyme kinetics:** Kinetics of enzyme action - Concept of E.S. complex, active site, specificity, derivation of Michaelis-Menten equation for uni-substratereactions.Differentplotsforthedeterminationof Km &Vmax and their physiological significances. Importance of Kcat/Km. Kinetics of zero & first order reactions. Significance and evaluation of energy of activation. Collision & transition state theories. Michaelis–pH functions & their significance. Classification of multi-substrate reactions with examples of each class. Reversible and irreversible inhibition. Competitive, non-and numerical basedon these. Suicide inhibitor.	12
III	**Mechanism of Enzyme Action:** Acid-base catalysis, covalent catalysis, proximity, orientation effect. Strain & distortion theory. Chemical modification of active site groups. Site directed mutagenesis of enzymes. Mechanism of action of chymotrypsin, lysozyme, glyceraldehyde 3-phosphate dehydrogenase, aldolase, carboxypeptidase, triosephosphate isomerase and alcohol dehydrogenase	12
IV	**Enzyme Regulation:** General mechanisms of enzyme regulation, product inhibition. Reversible (glutamine synthase & phosphorylase) and irreversible	12

	(proteases) covalent modifications of enzymes. Mono cyclic and multicyclic cascade systems with specific examples. Feedback inhibition and feed forward stimulation. Allosteric enzymes, qualitative description of "concerted"& "sequential"models forallosteric enzymes. Half site reactivity, Flip flop mechanism, positive and negative co-operativity with special reference to aspartate transcarbamoylase& phosphofructokinase	
Total		**48**

Text Books:

1. Fundamentals of Ezymology; 3rd Edn. Nicholas C. Price and Lewis Stevens, OxfordUniversity Press (2012).
2. Lehninger Principles of Biochemistry; D.L.Nelson and M.M. Cox, 7thEdn. MacMillanPublications(2017).

Reference Books:

1. An Introduction to Enzyme and Coenzyme Chemistry; Timothy B. Bugg,(1997)
2. Enzyme Kinetics; Roberts, D.V. (1977), Cambridge University Press.
3. The Enzymes; Boyer, 3rdedn. Academic Press, (1987).
4. Enzyme Kinetics; Irwin H. Segel (1976) Interscience-Wiley.
5. Enzyme Kinetics; the Steady state approach; Engel, P.C. (1981) 2nd Edn. Champmanand Hall.

PAPER II: MOLECULAR BIOLOGY
SUBJECT CODE: BCH154M202
CREDIT UNITS: L-T-P-C = 4-0-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
To educate students as well as to demonstrate knowledge and understanding of the molecular machinery of living cells.	Students will learn about – **CO1.** Students will learn about the central dogma of life, post transcriptional and post translational events, gene cloning and recombinant D.N.A. based technologies. **CO2.** Students will understand the processes of central dogma, proteins involved in post transcriptional and post translational events and principles guiding gene cloning and recombinant D.N.A. based technologies. **CO3.** Students will be able to analyse the interrelation of central dogma with the metabolic state of a cell, the implications of various processing events, and the nuances of gene cloning and recombinant D.N.A. based technologies.	Chalk and board teachings, power point presentations	Discussions, oral Questioning and analytical questions will be given to students. Internal exams and seminars will be conducted.

	CO4. Students will be able to evaluate the role of these events in maintaining a healthy cell.		

Detailed Syllabus:-

Modules	Course content	Periods
I	**Replication of Genetic Materials:** Nucleic acid as the genetic material (Griffith's experiment, Avery, MacLeod and McCarty's experiment, Hershey-Chase experiment), RNA vs DNA as genetic materials, Basic structure of DNA: Watson and Crick model of DNA structure, Renaturation and Denaturation of DNA, Packaging of DNA, Organelle DNA, Central Dogma of Molecular Biology, Features of DNA replication, Meselson and Stahl Experiments, Replication of DNA in prokaryotes and in Eukaryotes, Replication of Telomeric ends, Mitochondrial DNA replication	**12**
II	**Transcription and Translation:** Synthesis of RNA in prokaryotes and in eukaryotes, RNA polymerase and its promoters in prokaryotes and eukaryotes, 5' and 3'modification of mRNA in eukaryotes, Splicing, Alternative Splicing, RNA editing, Genetic codes, Translation in pro- and eukaryotes; post-translational protein modification, Protein Trafficking.	**12**
III	**Regulation of Gene Expression:** Regulation of gene expression in prokaryotes- Lac operon of *E. coli*, lactose as a carbon source in *E. coli*, molecular details of lac operon regulation, Trp operon in *E. coli*, regulation of the Trpoperon, regulation of gene expression in phage lambda.Regulation of Gene expression in eukaryotes- Transcription initiation by activators, transcription initiation by repressors, regulation of gene activity by histones and chromatin remodeling, gene silencing by DNA methylation and regulatory RNAs and libraries.	**12**
IV	**Recombinant DNA Technology:** DNA manipulating enzymes – restriction endonucleases; Restriction modification. Restriction analysis (shotgun cloning, RFLP, chromosome walking); Genetic finger printing; PCR – reverse and real time; Applied biotechnology: Molecular and antisense probes, micro & siRNA, gene silencing and its uses; Transgenics, Human genome; SNP; Functional genomics: concept and applications. Concept and applications of genome, transcriptome, proteome metabolome and CRISPR/Cas technology.	**12**
Total		**48**

Text Books

1. Lewin B. Genes IX (2011) & Genes XII (2017), Jones & Bartlett Publ.
2. Alberts B., Johnson A., Lewis J., Raft M., Robert K. and Walter P. (2015) Molecular Biology of the Cell, Sixth Edition, Garland Sci. Publ.
3. Glick B.R. and Pasternack J.J. (2009) Molecular Biotechnology, A.S.M. Press.

References Books

1. Lodish (2013) Molecular Cell Biology, WH Freeman & C.O.
2. Trop P.B.E. (2011) Molecular Biology Genes to Proteins, Jones & Bartlett Publisher.
3. Watson, J.D. et al. (2014) Molecular Biology of the Gene, 7th Ed., Benjamin Cummings Publ.
4. Hardin et al. (2015) Becker's World of the Cell, Ninth edition, Pearson, U. K.
5. Pollard et al. (2017) Cell Biology, Third Edition, Elsevier I.E.

PAPER III:IMMUNOLOGY
SUBJECT CODE: BCH154C203
CREDIT UNITS: L-T-P-C = 4-0-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
To introduce the students to the concept of Immunology and the functioning of the different components of the immune system.	Students will learn about **CO1.** Students will learn about organization of the immune system, antibodies and their diversity, immune reaction systems, immune disease, immuno-based assays and diseases associated with the immnune system. **CO2.** Students will understand the basis of immune system organization, working mechanism of the immune reaction systems, working principles of the immune-based assays and the molecular basis of immune system based diseases. **CO3.** Students will be able to analyse the inter-relations between the functioning of the different immune reaction systems. **CO4.** Students will be able to evaluate the role of different immune systems in the development of immune related diseases.	Chalk and board teachings, powerpoint presentations	Discussions, oral Questioning and analytical questions will be given to students. Internal exams and seminars will be conducted

Detailed Syllabus:-

Modules	Course content	Periods
I	**Introduction to Immune System**: Memory, specificity, diversity, innate and acquired immunity, self vs non-self-discrimination. Structure and functions of primary and secondary lymphoid organs. Cells Involved in ImmuneResponsesPhagocyticcellsandtheirkillingmechanisms.Tand B lymphocytes. Differentiation of stem cells and idiotypic variations. Nature of Antigen and Antibody. Antigen vs lmmunogen, HaptensStructure and functions of immunoglobulins. lsotypic, allotypic and idiotypicvariations.	**12**
II	**Generation of Diversity in Immune System:** Clonal selection theory - concept of antigen specific receptor. Organization and expression of immunoglobulin genes: generation of antibody diversity. T cellreceptor diversity. Humoral and Cell Mediated immune Responses. Kinetics of primary and secondary immune responses. Complement activation and its biological consequences. Antigen processing and presentation. Cytokines and co stimulatory molecules: Role in immune responses. T and B cell interactions. Major Histocompatibility Complex (MHC) Genes and Products. Polymorphism of MHC genes. Role of MHC antigens in immune responses. MHC antigens intransplantation.	**12**
III	**Applications of immune reactions:** Development, Regulation and Evolution of the Immune System Measurement of Antigen - Antibody Interaction. Production of polyclonal and monoclonal antibodies: Principles, techniques and applications. Agglutination and precipitation techniques. Radio immunoassay.Immunofluorescence assays: Fluorescence activated cell sorter (FACS) technique. Measurement of TCell activation.Fractionofleukocytesondensitygradient.Delayedtype hypersensitivity technique. Leukocyte migration inhibition technique.	**12**

	Cytotoxicity assay. Cytokines assay: ELISA and ELISPOT	
IV	**Immune associated diseases:** Tolerance vs Activation of immune System. Immune –tolerance. Immunosuppression. Hypersensitivity (Types I, II, III and IV).Immune Responses in Diseases. Immune responses to infectious diseases: viral, bacterial and protozoal, Cancer and immune system. Immunodeficiency disorders, Autoimmunity. Immunization, Active immunization (immunoprophylaxis). Passive immunization (Immunotherapy). Role of vaccines in the prevention of diseases.	**12**
Total		**48**

Text Books:

1. Kuby Immunology; Oven, Punt, Stranford, 8thEdn. W. H. Freeman (2018).
2. Roitt's Essential Immunology; Ivan, M. Rohitt&Petrer J Delves 13^{th} (2011) Blackwell Science.

Reference:

1. Antibodies– A Laboratory Manual; E. D. Harlow, David Lane, 2nd Edn. CSHLPress (2014).
2. Basic and Clinical Immunology; Stites et al., [6^{th}edn] (2011) Lange.
3. Veterinary Immunology: Ian R. Tizard, 10^{th}Edn. I.R. Thomsonpress.
4. The Immune System. By Peter Parham 4^{th}Edn. Publisher Garlandpublishing

PAPER IV: Practical on Enzymes, Molecular Biology & Immunology
SUBJECT CODE: BCH154M214
CREDIT UNITS: L-T-P-C = 0-0-8-4
STUDENT'S SCHEME OF EVALUATION: PRACTICAL (P)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
To introduce the basic techniques in molecular biology and spectrophotometric detection of biomolecules and their separation.	Students will learn about- **CO1.** Students will learn about the principles of enzyme based assays, estimation of haemoglobin and measurement of blood cells like WBC, R.B.C., separation of biomolecules using different techniques, designing primers, running PCR and immune assays. **CO2.** Students understand the basis of adding different chemicals and working steps for carrying out the experiments. **CO3.** Students will be able to analyse their results and make adjustments to their protocols when required. **CO4.** Students will be able to evaluate the results and discuss the reasons behind the results.	Chalk and board teachings, calculations guidance, video presentation of the methodology, conversion of moles, molar, mg, g, L etc, weighing, handling equipments and hands on practicals	Discussions, oral Questioning and analytical questions will be given to students. Students' hands on' learning will be regularly monitored

Detailed Syllabus:-

Modules	Course content	Periods
I	1. Isolation of protein and its estimation by Bradford method. 2. Assay of Vmax and Km of enzymes like salivary amylase and	**24**

	alkaline phosphatases. 3. Determination of effect of pH on salivary amylase.	
II	1. Blood grouping. 2. Enumeration ofWBC's 3. Radial immunodiffusion assay. 4. Double diffusion method.	**24**
III	1. Isolation of D.N.A. andAgarose gel electrophoresis. 2. Isolation of R.N.A. andAgarose gel electrophoresis. 3. Isolation of proteins and its separation using SDS PAGE.	**24**
IV	1. Demonstration of ELISA. 2. Demonstration of PCR 3. Designing of primers. 4. Demonstration of western blotting.	**24**
Total		**96**

Text Books:

1. Biochemistry, ed. Plummer Tata 3rdEdn.-McGraw Hill,
2. Practical Clinical Biochemistry, ed. Harold Varley, 4th edn. C.B.S. Publishers.

Reference Books:

1. Practical Clinical Biochemistry: Methods and Interpretation, 4thedn. Ranjna Chawla, Jaypee Brothers Medical Publishers.
2. Practical and Clinical Biochemistry for Medical Students, ed. T.N. Pattabhiraman, Gajanna Publishers.

3RD SEMESTER SYLLABUS

PAPER I: ADVANCED ENZYMOLOGY
SUBJECT CODE: BCH154M301
CREDIT UNITS: L-T-P-C = 4-0-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
To provide a deeper insight into the fundamentals of enzyme structure, multi-unit enzyme interactions and regulation.	**CO1.** Various aspects of enzymes like multisubstrate reactions, turnover of enzymes, cooperation between enzyme subunits, multienzyme systems, vitamins in enzyme reactions and reaction mechanisms of enzymes. **CO2.** Students will understand the biochemistry of multisubstrate reactions, cellular requirements for turnover of enzymes, conformational changes in protein cooperation, and advantages of multienzyme systems and role of vitamins in enzyme reactions. **CO3.** Students will be able to apply the gained knowledge in carrying out enzyme related studies. **CO4.** Students will be able to analyse the interdependence of the various aspects discussed in the functioning of the enzymes.	Powerpoint presentations, teaching using chalk and board, regular oral discussion sessions in the class on topics taught previously	Students will be asked questions; Quiz, internal assessment Tests will be conducted.

Detailed Syllabus:-

Modules	Course content	Periods
I	Review of unisubstrate enzyme kinetics and factors affecting the rates of enzyme catalyzed reactions. Michaelis pH functions and their significance. Classification of multisubstrate reactions with examples of each class.Kinetics of multisubstrate reactions. Derivation of the rate of expression for Ping Pong and ordered Bi Bi reaction mechanism. Concept of Convergent and Divergent evolution ofenzymes.Flexibility and conformational mobility of enzymes.	**12**
II	Enzymes Turnover and methods employed to measure Turnover of enzymes. Significance of enzymes Turnover. Protein - Ligand binding, including measurement, analysis of binding isotherms. Co- operativity phenomenon. Hill and Scatchard Plots. Allosteric enzymes, Sigmoidal kinetics, and their physiological significance. Symmetric and sequential modes for action of allosteric enzymes and their significance. Immobilized enzymes and their industrial applications.	**12**
III	Multienzyme system: Occurrence, isolation, and their properties. Polygenic nature of multienzyme systems. Mechanism of action and regulation of pyruvate dehydrogenase and fatty acid synthetase complexes. Immobilized MultienzymeSystems and their applications. Co-enzymes and cofactors: Water soluble vitamins as coenzymes. Metallo enzymes.	**12**
IV	Detailed Mechanisms of Catalysis of serine proteases. Ribonuclease, and Triose phosphate isomerases. Enzyme regulation: General mechanisms of enzyme regulation: Feed Back Inhibition and Feed forward stimulation; Enzyme repression, induction and degradation, control of enzymatic activity by products and substrates; Reversible and irreversible covalent	**12**

	modifications of enzymes.	
Total		**64**

Text books:

1. Enzymes; Trevor Palmer, East –2nd edn. West Press Pvt. Ltd., Delhi (2004).
2. Enzymes: A Practical Introduction to Structure, Mechanism, and Data Analysis; Robert A. Copeland, 2nd edn. Wiley-VCH Publishers (2000).Biochemical Calculations, Irwin H. Segel (1976) 2nd Ed. John Wiley and Sons.

Reference Books

1. Fundamentals of Ezymology; 3rd Edn. Nicholas C. Price and Lewis Stevens, Oxford University Press (2012).
2. *Lehninger Principles of Biochemistry*, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, U.S.A. 7th edition, 2017
3. Principles of Biochemistry; Smith et al., Ed. McGrawHill, (1986).
4. Introduction to Enzyme and Co-enzyme Chemistry. Ed. T. Bugg, 3rd edn. (2012), Blackwell Science.

PAPER II:CLINICAL BIOCHEMISTRY
SUBJECT CODE: BCH154M302
CREDIT UNITS: L-T-P-C = 4-0-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
To introduce the various parameters that determine a healthy and diseased state and to understand the workings of disease progression and development.	Students will learn about- **CO1.** The method of specimen collection and analysis; metabolic disorders, syndromes arising out metabolic disorders and the biochemistry of cancer. **CO2.** Students will understand the various aspects clinical biochemistry, their implications in health, causes and implications of metabolic disorders and the biochemical basis of cancer. **CO3.** Students can apply the knowledge gained to deliberate on clinical findings, check the relation of metabolites and syndromes associated with build-up of metabolites and the biochemical reasons leading to cancer progression. **CO4.** Students will be to analyse the relation of body metabolites and their deregulation with appearance of metabolic syndromes and relate the buildup of cancer specific conditions in the progression of cancer.	Teaching will be conducted both through white board mode and power point presentation Mode. Students will be asked to orally revise the previous class before every new class helping them in better understanding and their doubts cleared, if any.	Oral questions will be asked; students will be asked to discuss the topic. Quiz, internal assessment Tests will be conducted.

Detailed Syllabus:-

Modules	Course content	Periods
I	**Specimen collection and analysis:**Concepts of accuracy, Precision, Reliability, Reproducibility, Normal values, Specimen collection and	12

	processing, Blood Collection-Anticoagulants, Venipuncture, Urine collection, CSF, Aminotic fluid, pH of blood, acid base equilibrium, sodium, potassium, chloride, bicarbonate.	
II	**Inborn errors of metabolism:**Carbohydrate metabolism: galactosemia, Normal levels, renal threshold, Factors influencing blood glucose, Glycogen storage disorders, Pentosuria, Aminoacid metabolism-alkaptonuria, maple syrup urine disease, phenylketonuria, homocystinuria, proteinuria, albinism, multiple myeloma.Lipid metabolism: Hyperlipidemia,Hyperlipoproteinemia, Fatty liver.Nucleotide metabolism: Lesch-Nyhan Syndrome, Biochemistry of anemia, thalassemia, porphyria	12
III	**Endocrine Disorders:**Diabetes, obesity, atherosclerosis, Conn's syndrome, Addison's disease, Cushing's syndrome, hypo- and hyperthyroidism, gonadal dysfunction, dwarfism and gigantism,Diseases caused by chromosomal abnormalities-Down, Turner and Klinefelter syndromes.	12
IV	**Cancer Biochemistry:**carcinogenesis, characteristics of cancer cell,agents promoting carcinogenesis. Cellular differentiation, carcinogen, diagnosis of cancer, treatment of cancer.	12
Total		**48**

Text books:

1. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M. 7th Edition, 2017, WH Freeman and Company, New York, U.S.A.
2. A C Deb Fundamentals of Biochemistry, 10 edition, (2018), New Central Book Agency, London.

Reference Books:

1. Berg,J.M.,Tymoczko,J.L.andStryer.,Biochemistry,,W.HFreemanandCo.,9thedition,2019.
2. Allan Gaw, Michael Murphy, Rajeev Srivastava, Robert Cowan, Denis O'Reilly, Clinical Biochemistry, 5th Edition, 2013.
3. Text book of medical biochemistry by S.Ramakrishnan, K G Prasannan 3rdEdn.

Paper III: NEUROBIOCHEMISTRY
SUBJECT CODE: BCH154M303
CREDIT UNITS: L-T-P-C = 4-0-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
This paper provides a basic understanding of the nervous system and nerve cell functioning	**CO1.** Students will learn about the nervous system, transmissions, transmitters and neuro-degradative diseases. **CO2.** Students will understand the biochemistry of nerve signaling, transmitters and neuro-degradative diseases. **CO3.** Students will be able to analyse the functioning of the nervous systems and the significance the various transmitters in the nerve functioning.	Students will be asked to orally revise the previous class before every new class helping Them in betterunderstanding andtheirdoubtscleared, if any. Videos will beshown in the classfor a better Understanding ofthe concepts. Teaching will beconducted	Oral questions will be asked in the class. Problems will be assigned to test student's analytical ability. Class tests will be conducted for internal

	CO4. Students will be able to evaluate the biochemical events leading to the development of neuro-degradative diseases.	boththrough white boardmode and powerpoint presentations mode	assessment.

Detailed Syllabus:-

Modules	Course content	Periods
I	**Neuron:** Neuro cellular anatomy, neural membrane, classification of neuron, nerve fibers, axonal transport, neural growth, neuroglia, nervous system, blood-brain barrier, cerebrospinal fluid.	**12**
II	**Neuronal signalling:** Membrane potentials, ion channels, recording neuronal signals, ionic basis of resting potential and action potential, propagation of action potential. Synaptic transmission- Synapse, Electrical synapse transmission, chemical synaptic transmission, Synaptic transmitter release, synaptic potentials, synaptic delay, synaptic plasticity, molecular mechanism of synaptic transmission, myoneural junction.	**12**
III	**Neurotransmitters:** Chemistry, synthesis, storage, release, receptors and function- acetyl choline, catecholamines, serotonin, histamine, glutamate, asparatate, GABA, glycine, neuropeptides, nitric oxide.	**12**
IV	**Neural processing and neurodegenerative disorders:** Learning and memory, neurochemical basis of drug abuse, neurodegenerative disorders, Parkinson's disorder, Alzheimer's disorder, Amyotrophic Lateral Sclerosis, Senile Dementia	**12**
Total		**48**

Text Books:

1. Arthur C. Guyton and John E Hall, Text book of medical physiology 11th Edition;2006
2. David Nelson and Michael Cox, Lehninger Principles of Biochemistry, 4th edition;2005

Reference books:

1. Bruce Alberts, Alexander Johnson, Juliana Lewis, Martin Raff, Keith Roberts and Peter Walter, Molecular biology of the cell, 4th Edition;2004
2. Gordon Shepherd, Neurobiology, 3rd Edition;1994
3. MarkFBear,BarryWConnorsandMichaelAParadiso,Neuroscience:Exploringthebrain,4th Edition; 2015

Paper IV: CLINCAL BIOCHEMISTRY
SUBJECT CODE: BCH154M304
CREDIT UNITS: L-T-P-C = 4-0-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
To understand the basic nutrients and their functioning in the tissues	**CO1.** Students will be able to learn the basics of nutritional biochemistry, vitamins, minerals, and diseases associated with excessive sugar intake and malnutrition. **CO2.** Students will understand the role of different nutrients in maintaining the healthy state, and the working mechanisms of various nutrition associated diseases. **CO3.** Students will be able to analyse the biochemical functioning of different nutrients. **CO4.** Students will be able to evaluate the	Students will be asked to revise the previous class before every new Class orally. Teaching will be conducted boththrough whiteboard mode andpower pointpresentationMode.	Students will be asked questions, Quiz, internal assessment. Tests will be conducted.

	importance of nutrients in health and the issues arising out of malnutrition, fasting and obesity.		

Detailed Syllabus:-

Modules	Course content	Periods
I	Basic concepts: Function of nutrients. Measurement of the fuel values of foods. Direct and Indirect calorimetry. Basal metabolic rate: factors affecting B.M.R., measurement, and calculation ofB.M.R. Measurement of energy requirements. Specific dynamic action of proteins.Elements of nutrition: Dietary requirement of carbohydrates, lipids and proteins. Biological Value of proteins. Concept of protein quality. Protein sparing action of carbohydrates and fats. Essential amino acids, essential fatty acids, and their physiological functions.	**12**
II	Minerals: Nutritional significance of dietary calcium, phosphorus, magnesium, iron, iodine, zinc and copper.Vitamins: Dietary sources, biochemical functions, requirements, and deficiency diseases associated with vitamin B complex, C and A, D, E & K vitamins.	**12**
III	Malnutrition: Prevention of malnutrition, improvement of diets.Recommended dietary allowances, nutritive value of common foods. Protein-calorie malnutrition. Requirement of proteins and calories under different physiological states- infancy, childhood, adolescence, pregnancy, lactation and ageing.	**12**
IV	Starvation: Techniques for the study of starvation. Protein metabolism in prolonged fasting.Obesity: Definition, Genetic and environmental factors leading to obesity.	**12**
Total		**48**

Text Books:

1. Nutritional Biochemistry. Author, Tom Brody. Edition, 2. Publisher, Harcourt, Braces, 1999.
2. Principles of Nutritional Assessment (2005) Rosalind Gibson. Oxford university press.

Reference books:

1. Krause's Food and Nutrition Care process (2012); Mahan, L.K Strings, S.E, Raymond, J. Elsevier's Publications.
2. The vitamins, Fundamental Aspects in Nutrition and Health (2008); G.F. Coombs Jr. Elsevier's Publications.
3. Principles and Techniques of Practical Biochemistry, Keith Wilson and John Walker, 5thedition, 2000.
4. Physical Biochemistry, application to Biochemistry and Molecular Biology, David Freifelder, 2nd Edition, 1982.

PAPER V: Practical on Enzymes and Clinical Biochemistry SUBJECT CODE: BCH154C314 CREDIT UNITS: L-T-P-C = 0-0-8-4 STUDENT'S SCHEME OF EVALUATION: Practical Papers (P)

Facilitating the achievement of Course Learning Outcomes

Course Objective	**Course Learning Outcomes**	**Teaching and Learning Activity**	**Assessment Tasks**
To develop skills of performing	Students will learn- **CO1.** Students will learn the principles of determination of activation energy, estimation of	Chalk and board teachings, calculations	Discussions, oral Questioning and

basic biochemical tests important in clinical investigations.	serum metabolites like cholesterol, urea etc, activity determination of SGOT and SGPT. **CO2**. Students will understand the principles and the various reasons behind the protocols being followed. **CO3.** Students will be able to analyse the results and make changes in the protocols as required. **CO4.** Students will be able to evaluate the health status based on the results of the serum metabolites and enzymes.	guidance, video presentation of the methology, conversion of moles, molar, mg, g, L etc, weighing, handling equipments and hands on practicals	analytical questions will be given to students. Students' hands on'learning will be regularly monitored

Detailed Syllabus:-

Modules	Course content	Periods
I	**1.** Determination of activation energy of enzyme. **2.** Determination of influence of temperature on enzyme activity. **3.** Immobilisation of urease in calcium alginate beads.	24
II	**4.** Determination of type of inhibition (reversible or irreversible) of amylase. **5.** Modelling of an enzyme active site using in silico method.	24
III	**6.** Estimation of glucose by Folin-Wu method. **7.** Estimation of cholesterol by Zack's method. **8.** Estimation of haemoglobin by Wong's method **9.** Estimation of urea in blood by Diacetylmonoxime method. **10.** Estimation of serum calcium by Clark and Collips method.	24
IV	**11.** Determination of A/G ratio by Biuret method. **12.** Analysis of SGOT-SGPT (A.S.T., A.L.T.) / creatine kinase/ acid or alkaline phosphatase. **13.** Qualitative analysis of Urine sample for normal and abnormal constituents. **14.** Estimation of uric acid in serum and urine by Caraway's method **15.** Determination of urine Chloride by Volhard-Arnold method. **16.** Estimation of urine Bilirubin.	24
Total		**96**

Text books:

1. Biochemistry, ed. Plummer Tata 3rd Edn.-McGraw Hill,.
2. Practical Clinical Biochemistry, ed. Harold Varley, 4th edn. C.B.S. Publishers.

Reference books:

1. Hawk's Physiological Chemistry, ed. Oser, 14th Edn., Tata-McGraw Hill.

2 .Practical Clinical Biochemistry: Methods and Interpretation, 4th edn. Ranjna Chawla, Jaypee Brothers Medical Publishers.

3. Practical and Clinical Biochemistry for Medical Students, ed. T.N. Pattabhiraman, Gajanna Publishers.

4^{TH} SEMESTER SYLLABUS
PAPER I: MICROBIOLOGY SUBJECT CODE: BCH154M401 CREDIT UNITS: L-T-P-C = 4-0-0-4 STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
The course aims To give students a conceptual foundation for understanding pathogenic microorganisms, with a focus on the underlying processes of their pathogenicity.	**CO1.** Study the distinction between prokaryotes and eukaryotes **CO2.** Gain an understanding of microbial diversity, bacterial morphology and its growth patterns. **CO3.** Study about microbe pathogenicity and analyze the mechanism of action and epidemiology of pathogens, **CO4.** As well as evaluate the industrial applications of microorganisms.	Students will be asked to orally revise the previousclass before every new class helping them in better understanding andtheir doubtsCleared, if any.Videos will beshown in the classfor a betterunderstanding ofThe concepts.Teaching will beconducted boththrough white board mode and power point presentationsmode	Oral questions will be asked inthe class. Problems will be assigned to test student'sanalytical ability. Class tests willbe conducted for internalAssessment.

Detailed Syllabus:-

Modules	Course content	Periods
I	**Introduction to Microbiology and Diversity:**Contributions of Anton von Leeuwenhoek, Louis Pasteur, Robert Koch, Joseph Lister, Alexander Fleming, Development of the field of soil microbiology: Contributions of Martinus W. Beijerinck, Sergei N. Winogradsky, Selman A. Waksman, Paul Ehrlich, Elie Metchnikoff and Edward Jenner. Systems of classification (Binomial Nomenclature, Whittaker's five kingdom and Carl Woese's three kingdom classification systems and their utility). Difference between prokaryotic and eukaryotic microorganisms. General characteristics of different groups: Acellular microorganisms (Viruses, Bacteriophages, Viroids, Prions) and Cellular microorganisms (Bacteria, Algae, Fungi and Protozoa) with emphasis on distribution and occurrence, morphology, mode of reproduction and economic importance.	12
II	**Microbial genetics, Physiology and Metabolism:**Genome organization. Coli, Saccharomyces.Organelle genome: Chloroplast and Mitochondria. Types of plasmids – F plasmid, R Plasmids, colicinogenic plasmids, Ti plasmids, linear plasmids, yeast-2 μ plasmid.Transformation, Conjugation (mechanism, Hfr and F' strains, Interrupted mating technique and time of entry mapping). Transduction (Generalized transduction, specialized transduction). Definitions of growth, Batch culture, Continuous culture, generation time and specific growth rate. Nutritional categories and requirements of microorganisms.Concept of aerobic respiration, anaerobic respiration and fermentation.	12
III	**Industrial and Medical Microbiology:**Microorganisms in industries – citric acid and lactic acid production, production of beer and wine.Food borne infection and food borne intoxication.Epidemiology of bacterial,	12

	fungal and viral diseases, production of antibiotics from microorganisms and chemotherapy.	
IV	**Tools and Techniques in Microbiology:**Culture media: components of media, natural and synthetic media, chemically defined media, complex media, selective, differential, indicator, enriched and enrichment media. Pure culture isolation: Streaking, serial dilution and plating methods; cultivation, maintenance and preservation/stocking of pure cultures; Stain and staIning techniques.Microscopy and its different types.Asexual methods of reproduction, logarithmic representation of bacterial populations, phases of growth, calculation of generation time and specific growth rate.Physical and Chemical methods of microbial control: Types and mode of action.	12
Total		**48**

Text books:

1. *Lehninger Principles of Biochemistry*, Nelson, D.L., Cox, M.M., WH Freeman andCompany, New York, USA. 7th Edition, 2017
2. Prescott, Harley and Klein's Microbiology, J.M. Willey, L.M. Sherwood and C.J.Woolverdon (2017), 10th ed.,McGraw Hill Publishers.
3. Frazier and Westhoff. Food Microbiology 5th Edition, McGrawHill

Reference Books:

1. Brock et al. Jawetz, Milnick and Adelberg's Medical Microbiology. 27th Edn (2016) Lange Me
2. Biology of Microorganisms, 15th Edn. Brock Prentice Hall(2017).
3. Industrial Microbiology; Miller and Litsky (Eds.) (1976) McGraw HillPublishers.
4. Microbiology; Lansing M. Prescot, Hartley and Klein, 5th Edn. McGraw Hill(2002).
5. Microbiology; Essentials and Applications, Larry Mckane and J. Kandel (19)2nd Edn.McGrawHillpublishers.

PAPER II: MEMBRANE BIOLOGY
SUBJECT CODE: BCH154M402
CREDIT UNITS: L-T-P-C = 4-0-0-4
STUDENT'S SCHEME OF EVALUATION: Theory Papers (T)

Facilitating the achievement of Course Learning Outcomes

Course Objective	Course Learning Outcomes	Teaching and Learning Activity	Assessment Tasks
The course aims To educate students about the structures and functions of basic components of prokaryotic and eukaryotic cells.	Students will **CO1.** Study about the structure of bio-membranes, the fluid mosaic model, and the liposome. **CO2.** Micelles, membrane asymmetry, macro and micro domains in membranes, lipid rafts, caveolae, tight junctions, and R.B.C. membrane architecture will be discussed. **CO3.**Explore membrane dynamics and the membrane transport thermodynamics, **CO4.** As well as technical applications for studying membrane dynamics.	Students will be asked to orally revise the previous class before every new class. Teaching will be conducted both through white board mode and power point presentations mode	Oral questions will be asked inthe class. Problems will be assigned to test student's Analytical ability. Class tests willbe conducted for internalassessment.

Detailed Syllabus:-

Modules	Course content	Periods
I	**Introduction to bio membranes:** Composition of bio membranes - prokaryotic, eukaryotic, neuronal, and subcellular membranes.Study of membrane proteins. Fluid mosaic model with experimental proof. Monolayer, planer bilayer and liposomes as model membrane systems.	**12**
II	**Membrane structures:** Polymorphic structures of amphiphilic molecules in aqueous solutions - micelles and bilayers. C.M.C., critical packing parameter. Membrane asymmetry. Macro and micro domains in membranes. Membrane skeleton, lipid rafts, caveolae and tight junctions. R.B.C. membrane architecture.	**12**
III	**Membrane dynamics:** Lateral, transverse, and rotational motion of lipids and proteins. Techniques used to study membrane dynamics- FRAP, TNBS labelling etc. Transition studies of lipid bilayer,Transition temperature. Membrane fluidity, factors affecting membrane fluidity.	**12**
IV	**Membrane transport:** Thermodynamics of transport. Simple diffusion and facilitated diffusion. Passive transport - glucose transporter, anion transporter and porins. Primary active transporters - P type ATPases, V type ATPases, F type ATPases. Secondary active transporters - lactose permease, Na+ -glucose symporter. A.B.C. family of transporters - M.D.R., CFTR.Group translocation. Ion channels - voltage-gated ion channels (Na+ /K+ voltage-gated channel), ligand-gated ion channels (acetyl choline receptor), aquaporins, and bacteriorhodopsin. Ionophores - valinomycin, gramicidin.	**12**
Total		**48**

Text books:

1. *Lehninger Principles of Biochemistry*, Nelson, D.L., Cox, M.M., WH Freeman andCompany, New York, U.S.A. 7th edition, 2017
2. Victor Rodwell, David Bender, P. Anthony Weil, Peter Kennelly. *Harpers Illustrated Biochemistry* 31th Edition,2018
3. Text book of medical biochemistry by M N Chatterjee, Rana Sindhe. 8thEdn
4. Text book of medical biochemistry by S. Ramakrishnan, K G Prasannan. 3rdEdn.

Reference Books:

1. J.M. Berg, J.L. Tymoczko, L. Stryer. . Biochemistry, 9th Edn. (2019) WH Freeman and Company, New York and England.
2. R. Verna. Membrane Technology, Raven Press, New York., U.S.A.
3. H. Lodish, A. Berk, S.L. Zipursky, P. Matsudaira, D. Baltimore, J. Darnell. Molecular Cell Biology, 8th Edn. WH Freeman and Company, NY and England
4. H.R. Petty. Molecular Biology of Membranes Structure and Function, Plenum Press, New York, U.S.A. and London.
5. D.F.H. Wallach. Membrane Molecular Biology of Neoplastic Cells, (1975) Elsevier Scientific Publishing Company, Amsterdam, Oxford and New York., U.S.A.